Practice Question

Q.1

```
  ABCD0
   2262
 ------
  ADF32
```

∴ a = 61h

```
val1  byte  10h
val2  dw    5566h
val3  db    'abcd', 0
val4  db    11h, 22h, 33h
val5  dw    50h, 60h, 70h, 80h
val6  db    ?, ?
val7  db    4 DUP (?)
val8  dw    3DUP (55h), 56h, 57h
```

Address	Value	Variable	Add	Value	Variable
ADF32	10	val1	4E	00	
33	66	val2	4F	55	
34	55		50	00	
35	61	val3	51	56	
36	62		52	00	
37	63		53	57	
38	64		54	00	
39	0		55		
3A	11	val4	56		
3B	22		57		
3C	33		58		
3D	50	val5	59		
3E	00		5A		
3F	60		5B		
40	00		5C		
41	70		5D		
42	00		5E		
43	80		5F		
44	00				
45	?	val6			
46	?				
47	?	val7			
48	?				
49	?				
4A	?				
4B	55	val8			
4C	00				
4D	55				

.data

quard	Qword	'ABCD', 0ABCDEFABCDh
doubleWord	dd	'ABC', 01234ABCDh
my world	word	'AB', 'CD', 0ABCDh
my byte	db	'ABCD', ABh, 20h

A = 41h

For byte type "ABCD" is an array and store in memory start from A → B → C → D

For Remaing word, dword, qword it start from right D → C – B → A

Note:
For word → 2 Charactr or less
dword → 4 or less
qword = 8 or less
(But not more than)

Address	Value	Variable	Address	Value	Variable
04010	44	quard	04024	CD	
4011	43		25	AB	
12	42		26	~~AB~~34	
13	41		27	12	
14	00		28	42	myword
15	00		29	41	
16	00		30	24	
17	00		31	43	
18	CD	~~doubleword~~	32	CD	
19	AB		33	AB	
1A	CD		34	41	mybyte
1B	AB		35	42	
1C	00		36	43	
1D	00		37	44	
1E	00		38	AB	
1F	00		39	20	
20	43	doubleword	3A		
21	42				
22	41				
23	00				

Q.3

```
  35AD0
   1002
  -----
  36AD2
```

```
.data
Val1 db 10h
Val2 dw 01234h
Val3 dword 11223344h
Val4 db 'a'
Val5 dw 34h
Val6 db 'ABCD', 0
Val7 db 10h, 20, 30h, 40h,
Val8 dw 70h, 80h, 90h, 100h, 110h
Val9 db 3 DUP(5h)

.code
  mov Al, val1 ; AL = 10
  mov al, [val1+2] ; AL = 12
  mov ax, val2 ; Ax = 1234
X mov eax, [val3+2] ; eax = 34611122h
X mov al, val5 ; al = Error
  mov ah, [val6+2] ; ah = 43h
  mov bl, [val7+1] ; bl = 20h
X mov cx, [val8+3] ; Cx = 9000h
X mov al, [val9+3] ; AL = Garbage.
```

(correct value is 11223344h) not correct

missmatch variable

correct value is 0090 So its incorrect

Store garbage value

Addre	Value	Add	Value
36AD2	10 val1	EE	05 val9
D3	34 val2	EF	05
D4	12 val3	F0	05
D5	44	F1	
D6	33	F2	
D7	22	F3	
D8	11 val4		
D9	61 val5		
DA	34 val6		
DB	00		
DC	41		
DD	42		
DE	43		
DF	44 val7		
E0	10		
E1	20		
E2	30		
E3	40 val8		
E4	70		
E5	00		
E6	80		
E7	00		
E8	90		
E9	00		
EA	00		
EB	01		
EC	10		
ED	01		

Q.4

A5A20
1002
A6A22

-4 → 00000100
11111100
F C

-3 = FE

```
.data
1 mybyte db 10h, 'A', 11001101b, 40h
2 myword dw 1Ah, 3Bh, 72h, 44h, 66h
3 mydouble dd 1,2,3,4,5
  varD LABEL Dword
  varW LABEL WORD
4 myByte3 SBYTE -4,-2,2,1
                36h 43h 50h
5 myByte2 Byte 54,67,80,0ABh
6 myword2 word 3 DUP(?), 2000h
.code
mov esi, offset mybyte ;
mov al, [esi]          ; al= 10
mov esi, offset mydouble
mov al, [esi-3]        ; al=00
mov esi, offset myword +2
mov ax, [esi]          ; ax=003B
mov edi, 8
mov esi, 2
mov edx, [varD+esi]    ; edx=43360102
mov edx, Mydouble[edi] ; edx= 00000003
mov ax, word PTR [mybytes2+7]
                       ; ax = 5043
```

Address	Value	Address	Value
A6A23	10 1	3F	00
24	41	40	00
25	CD	41	05
26	40	42	00
27	1A 2	43	00
28	00	44	00
29	3B	45	FC 4
2A	00	46	FE
2B	72	47	02
2C	00	48	01
2D	44	49	36 5
2E	00	4A	43
2F	66	4B	50
30	00	36	AB
31	01 3	4B	? 6
32	00	4E	?
33	00	4F	?
34	00	50	?
35	02	51	?
36	00	52	?
37	00	53	00
38	00	54	20
39	03	55	
3A	00		
3B	00		
3C	00		
3D	04		
3E	00		

Q 5

0DDh & 0C6h

	OF	CF	AC	SF	ZF	PF
1	0	1	1	1	0	1
2	0	1	0	1	0	1
3	1	1	1	0	0	1

1

  0D
  C6
1 A 3

2 145D − 73AB

borrow +16 +16
CF=1 145D small
− 73AB − big
A0B2

0 0110010 PF=1

OR

16's Comp of +73AB

FFFF
− 73AB
8C54
+1
$\boxed{8C55}$ 16's Comp

145D
8C55
A0B2

3 8F
A1
1 3 0
CF
0011 0000 PF=1

Q.6

1 0 − Ax

Borrow 000 0 +16
CF=1 FFFF
0001 = Ax

2 0001
+ 0001
0002 = Ax

3 B 0001 +16
0002
FFFF = BX
$\boxed{1111\ 1111}$ PF = 1

4 Ax − Bx
B → 0002 +16
FFFF
0003
0000 0011 PF = 1

5 0003
+ 1
0004

No update any flag
So all flags is zero

For Inc Dec. both CF not effect

		OF	CF	AC	SF	ZF	PF
0001	1	0	1	1	0	0	0
0002	2	0	0	0	0	0	0
FFFF	3	0	1	1	1	0	1
0003	4	0	1	1	0	0	1
	5	0	0	0	0	0	0

Q.5: Write values of status flag (carray , sign, Auxilary carry, parity , and zero)for theeach of the following instructions:

Instruction	OF	CF	AC	SF	ZF	PF

MOV al, 0DDh
MOV bl, 0C6h
ADD al, bl
	0	1	1	1	0	1

MOV ax , 145Dh
MOV bx, 73ABh
SUB ax,bx
	0	1	0	1	0	1

MOV ax, 0AB8Fh
MOV bl,0A1h
ADD al,bl
	1	1	1	0	0	1

Q.6: Given AX= 0FFFFh , BX = 0001h . Determine the values of status flag (carray , sign, Auxilary carry, parity , and zero)for following cases:

Instruction	OF	CF	AC	SF	ZF	PF
NEG AX	0	1	1	0	0	0
ADD AX, BX	0	0	0	0	0	0
SUB BX,AX	0	1	1	1	0	1
SUB AX, BX	0	1	1	0	0	1
INC AX	0	0	0	0	0	0

Q.7: Find the value of registers in each instruction :

```
.data
arrB DB 11h, 22h, 33h, 44h, 55h
.code
mov esi, OFFSET arrB
mov al, [esi]             ; ?
mov ax, WORD PTR [esi]    ; ?
mov ax, WORD PTR [esi+1]; ?
```

al = 11h
ax = 2211h
ax = 3322h

Q.8: Find the value of registers in each instruction :

```
.data
arrD DD 12345678h
.code
mov esi, OFFSET arrD
mov ax, WORD PTR [esi]      ; ?
mov ax, WORD PTR [esi+2]    ; ?
mov al, BYTE PTR [esi+3]    ; ?
```

ax = 5678h
ax = 1234h
al = 12h

Q.9: Find the value of registers in each instruction :

```
.data
mix DB 10h
    WORD 2000h
    DB 30h
    WORD 4000h
.code
mov esi, OFFSET mix
mov al, [esi]              ; ?
mov ax, WORD PTR [esi+1] ; ?
mov al, [esi+3]            ; ?
mov ax, WORD PTR [esi+4] ; ?
```

al = 10h
ax = 2000h
al = 30h
ax = 4000h

Q.10: Find the value of registers in each instruction :

```
.data
arr DB 0AAh, 0BBh, 0CCh, 0DDh, 0EEh
.code
mov esi, OFFSET arr
mov al, [esi]                 ; ?
mov ax, WORD PTR [esi+1]     ; ?
mov ax, WORD PTR [esi+2]     ; ?
mov al, [esi+4]               ; ?
```

al = AAh
ax = CCBBh
ax = DDCCh
al = EEh